Contents

1 Overview 2

2 Product Opportunity 2
- 2.1 Physical Strain 2
- 2.2 Cognitive Strain 2
- 2.3 Social and Emotional Well-being 2

3 Concept Development Approach 2
- 3.1 Initial Ideation 2
- 3.2 Selection Rationale 2
- 3.3 Original Design Issues 3
  - 3.3.1 Form and User Interface 3
  - 3.3.2 Pot Attachment 3
  - 3.3.3 Stirring Attachment 3

4 Prototyping/Testing 3
- 4.1 CAD Prototyping - Digital Prototyping 3
  - 4.1.1 Different Versions 3
- 4.2 Rapid Prototyping 4
  - 4.2.1 Form Prototyping 4
  - 4.2.2 Pot Attachment 5
  - 4.2.3 Stirring Attachment 5
  - 4.2.4 Central Rod Attachment 5
  - 4.2.5 Planetary Gears 6
- 4.3 Material selection 6

5 Electronic Component Selection 6
- 5.1 Outputs 6
  - 5.1.1 Motor Requirements 6
  - 5.1.2 Neopixel Ring 7
  - 5.1.3 Speaker 7
- 5.2 Inputs 7
  - 5.2.1 Rotatory Encoder 7
  - 5.2.2 Buttons 7
  - 5.2.3 Infrared Sensor 7
- 5.3 Other Components 7
- 5.4 Battery Calculations 7
- 5.5 Electronic Circuit Design 8

6 Design For Assembly 8
- 6.1 Snap fittings 8
- 6.2 Screws 9
- 6.3 Provisioning For Assembly 9

7 Design For Manufacturing 9
- 7.1 Injection Moulding 9
- 7.2 Stamping 10
- 7.3 Extrusion 10

8 Branding Consideration 10
- 8.1 Benchmarking Analysis: Product 10
  - 8.1.1 Uutensil Stirr Automatic Pan Stirrer 10
  - 8.1.2 Gourmia Stirrific 10
  - 8.1.3 Ardente Gourmet Stirrer 11
  - 8.1.4 StirMate Smart Pot Stirrer 11
- 8.2 Benchmarking Analysis: Brand 11
  - 8.2.1 Kitchenaid 11
  - 8.2.2 Joseph Joseph 11
  - 8.2.3 Ototo 11
  - 8.2.4 Peleg Design 11
- 8.3 Brand Selection 11

9 Brand Implementations 11
- 9.1 Brand Implementations on the Physical Product 11
  - 9.1.1 Functional Aesthetics 11
  - 9.1.2 Humour and Whimsy 11
- 9.2 Brand Implementations on the Product's Values 12
  - 9.2.1 Functionality 12
  - 9.2.2 Aesthetics 12
  - 9.2.3 Humour 12

10 Product Compliance: Research 12
- 10.1 UKCA 12
- 10.2 CE 12
- 10.3 Marking Requirements 12
- 10.4 General Legislation 12

11 Product Compliance: Conformity 13
- 11.1 RoHS (Restriction of Hazardous Substances in Electrical and Electronic Equipment) 13
- 11.2 EMC/EMF (Electromagnetic Compatibility and Electromagnetic Fields) 13
- 11.3 WEEE (Waste from Electrical and Electronic Equipment) 13
- 11.4 EcoDesign Directive 14
- 11.5 Battery Directive 14
- 11.6 Food Contact Materials Regulation 14
- 11.7 Declaration 14

12 Labelling and User Guide 14
- 12.1 Product Labels 14
  - 12.1.1 WEEE Label 14
  - 12.1.2 UKCA and CE Labels 14
  - 12.1.3 Food Contact Material Label 14
- 12.2 User Guide 15
  - 12.2.1 Front and Back Cover 15
  - 12.2.2 Instructions 15
  - 12.2.3 Attachment Shapes 15
  - 12.2.4 Safety 15
  - 12.2.5 Maintenance and Support 15
  - 12.2.6 Declaration of Conformity 16

13 Product Packaging Design 16
- 13.1 Branding and Inspiration 16
- 13.2 Target Market 16
- 13.3 Functionality and Design 16
- 13.4 Internal Design 16
- 13.5 Prototyping 17
- 13.6 Labels 17
  - 13.6.1 The Mobius Loop (20 PAP) 17
- 13.7 Material Choice 17

14 Final Design - Features 17
- 14.1 Tactile Rotary Control 17
- 14.2 Visual and Auditory Feedback 18
- 14.3 Adjustable Sizes 18
- 14.4 Mouldable Stirrer 18

15 Business Case 18
- 15.1 The Market 18
- 15.2 Product Feasibility, Desirability And Viability 18
- 15.3 Revenue Model 19

16 Object Validation 19
- 16.1 Mechanical & Electronic Functions 19
  - 16.1.1 Controlling the device 19
  - 16.1.2 Adjustable Dimensions 19
  - 16.1.3 Gripping the pot 19
- 16.2 Feedback 19
- 16.3 Elena Sachetti 19
- 16.4 Anthony Walsh 19

17 Project Plan Gantt Chart 19

18 Team Roles & Contributions 20
- 18.1 Emmanuel - Chief Information Officer 20
- 18.2 Sinai- Chief Operations Officer 20
- 18.3 Francesco - Chief Technical Officer 20
- 18.4 Alex - Chief Creative Officer 20

19 References 21

1 Overview

"Cooking is like love. It should be entered into with abandon or not at all." Harriet Van Horne. This quote captures the essence of cooking—not just as a routine task but as an expression of passion and enjoyment. However, as individuals age, the complexities of cooking can turn this once joyful activity into an arduous chore. For many elderly individuals, the physical limitations and cognitive challenges that come with aging can dampen their enthusiasm for cooking, which can have a major impact on their health and well-being.

2 Product Opportunity

In order to make cooking more accessible and enjoyable for all, our project aims to harness innovative technology to assist the elderly with varied physical capabilities. From a combination of primary and desk research we discovered the following key avenues for exploration.

2.1 Physical Strain

The primary aim of our product is a kitchen tool that reduces the physical effort required for elderly people when cooking. Common challenges such as reduced grip strength and more specifically, difficulty lifting and stirring as a result of this.

2.2 Cognitive Strain

Forgetfulness can complicate daily tasks such as cooking and shopping for the elderly, posing safety risks and causing frustration. It is essential for our product to integrate safety features and reminders to help reduce cognitive load whilst maintain the enjoyment of cooking.

2.3 Social and Emotional Well-being

Recognising cooking as a chance for social interaction, our aim is to create a solution that not only simplifies the cooking process but also enables elderly individuals to host and engage in larger family meals and gatherings. Seniors should be afforded the opportunity to spend more meaningful time with family and friends, and let cooking be transformed from a solitary task into an enjoyable and socially enriching activity.

3 Concept Development Approach

3.1 Initial Ideation

As a team, we explored various solutions to ensure they aligned with the key avenues we aimed to address. This process led each of us to propose one finalised solution:

- **CQK**: The Culinary Quest Knife is a state-of-the-art knife with motorised slicing and an app for social sharing, recipe discovery, and progress tracking, making cooking enjoyable and connected.
- **StirMaster Pro**: The StirMaster Pro offers hands-free stirring, ensuring even heat and texture. It fits various pot sizes, features durable, food-safe materials, and has variable speed settings for all dishes.
- **Mashomatic**: With a powerful motor and spring suspension, the Mashomatic effortlessly mashes soft fruits and vegetables in seconds.
- **Legacy**: Legacy is a smart scale that records traditional family recipes and provides visual and vocal instructions, making it easy to recreate family favorites.



Figure 1. Finalised Ideation

3.2 Selection Rationale

Having conducted a range of user feedback going back to our original interviewees we highlighted a range of issues with our original designs.

- **CQK**: Despite being highly-functional and greatly reducing physical strain on the user, it may prove dangerous in manufacture. Additionally, it may prove difficult to hold securely with one hand in operation without sizeable dampening to restrict vibrations in the handle.
- **Mashomatic**: With the strength of the motor, the same issues may be encountered as with the knife. Moreover, the product boasts a very narrow scope of applications.
- **Legacy**: Legacy is a great product which many users would appreciate, but may not fit the brief of a hand-held electromechanical product best.
- **StirMaster Pro**: The automatic pot stirrer bridges the gap between meeting the brief, fulfilling a wide range of applications for a variety of meals, whilst easing physical and cognitive strain from stirring and remembering timings of food on the stove. Furthermore, it provides a smooth and stress-free cooking experience for elderly users.

3.3 Original Design Issues

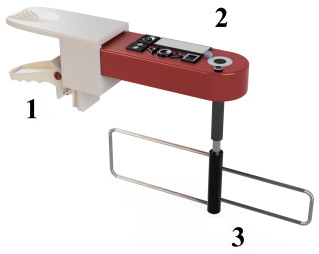


Figure 2. Original Design

3.3.1 Form and User Interface

The form of the product with the placement of the screen and buttons over the centre of a hot pot may be uncomfortable for the user during setup. Moreover the screen may prove both difficult to view and experience malfunctions with the combination of steam and heat.

Then taking inspiration from a watch face, the interface was adapted to operated on a rotary encoded gear such that the outer ring of the central hub dependent on which mode was pressed would be able to regulate speed and duration of stirring. This paired visual feedback of an LED ring provided a much simpler, elegant and effective interface for our target market to operate.

3.3.2 Pot Attachment

The one-sided pot attachment bodes well for inclusivity catering towards those with physical handicaps requiring one hand for operation. In practice, the squeezing of the handle may prove tiresome on the users forearm and multiple points of contact between the device and the pot may provide more favourable security.

3.3.3 Stirring Attachment

The attachment is secured by a bayonet mechanism and would prove effective. Multiple attachment would allow for an each pot size and and food consistency to be catered for. Although, it may be cumbersome to store and could benefit from exploration of ways to avoid this.

4 Prototyping/Testing

Prototyping is a critical phase of our project, undertaken to ensure that all components and planned features function as intended. This phase involved several stages, from CAD modeling to physical testing, to refine our automatic stirrer design for the elderly.

4.1 CAD Prototyping - Digital Prototyping

The initial stage of prototyping involved using CAD software to visualise different concepts. This allowed us to create multiple iterations of models and evaluate their feasibility without incurring material costs and time wastage due to manufacture. During this stage, we determined that incorporating a screen would lead to a dial mechanism that was too bulky, and we were concerned about the potential impact of heat on the screen. This led us to the decision of using an LED ring as the primary method of communication.

4.1.1 Different Versions

There were 5 distinct versions to the design:

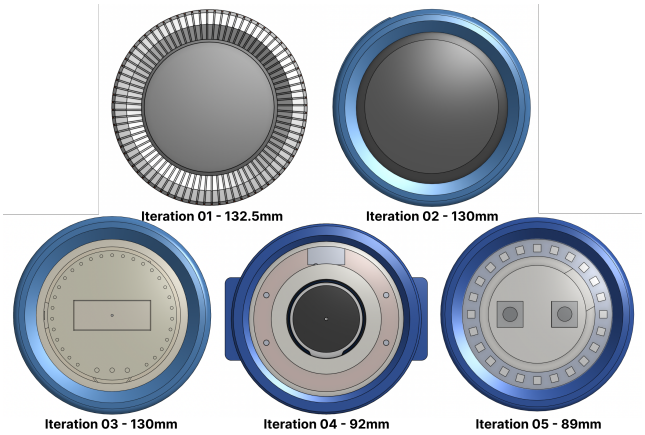


Figure 3. Screen Display Iterations

- **Version 1:** This version was to see the design with a circular screen, and in the style of KitchenAid, in order to start having an idea of what components we wanted to have, dimensions, and if we liked the look of the brand.
- **Version 2:** From the previous Version, we considered changing our brand to JosephJoseph in order to give our product a more playful aesthetic. This version still included the screen.
- **Version 3:** Here, after collecting user feedback, we decided to remove the screen and replace it with a custom PCB which including a ring of LEDs and segmented display. However, this proved too complicated.
- **Version 4:** This was the first version including the outside shell as we wanted to experiment with having a flexible attachment and supports, and an LED ring with a small screen. This design did not receive good feedback, but from this, the cuttlefish influence came through.
- **Version 5:** The final design made a concerted effort to resemble a cuttlefish. This required us to rebrand our company to PelegDesign in order to better reflect the biomimicry of the design, which is demonstrated by the stirring attachment and support arms that resemble tentacles, the handles that resemble squid wings, and the two eyes that give the appearance that our product is peeking out of the pot. A mouldable attachment was also included, with the soul visual output kept being the LED ring.

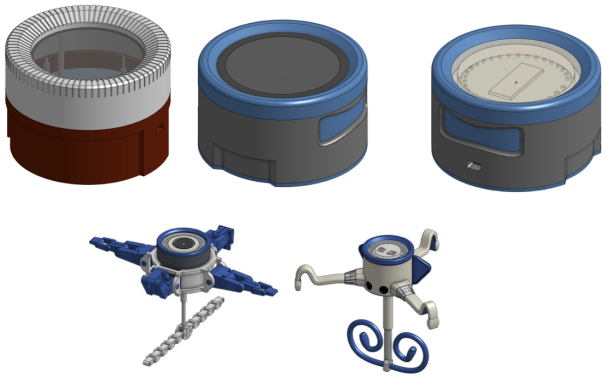


Figure 4. 5 Versions Iterations

4.2 Rapid Prototyping

4.2.1 Form Prototyping

Based on our CAD designs, we 3D printed several models to assess their form and usability. This process validated our decision to omit screens, as they were not ideal for the dial mechanism, which would have been too large. This can be seen in Figure 4.

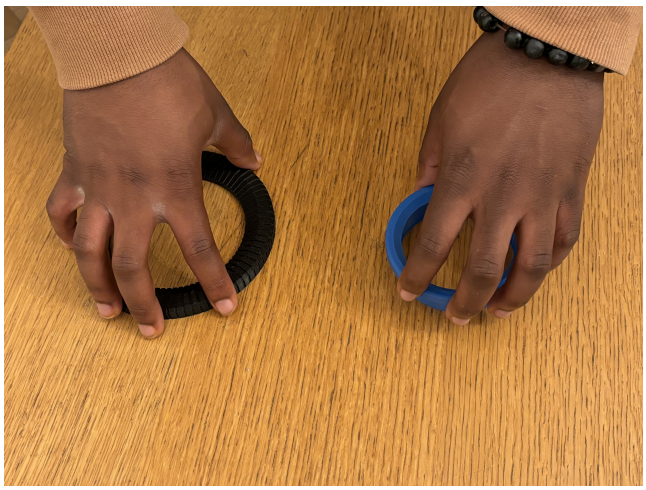


Figure 5. Screen Display Iterations

Iteration 5 was selected as the design to move forward with. To validate this choice, we 3D printed the hub and its casing to check the initial form and fit. The inner casing required two iterations: the first prototype ensured that the electronics fit correctly, while the second prototype added snap-fit gear pins to secure the gears within the casing. Additionally, the base was modified with three flattened edges to prevent rotation during motor operation, and six stubs

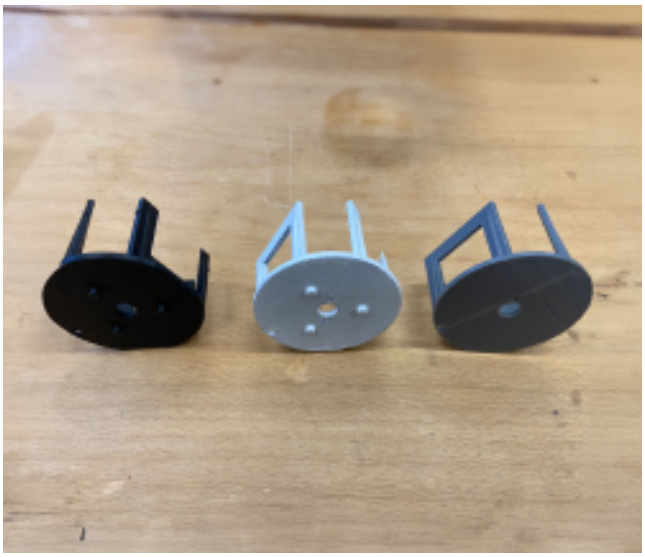


Figure 6. Inner Shell Iterations

were added to the top of the inner casing to facilitate knob rotation.

For the outer casing, three iterations were necessary. The initial iteration revealed difficulty in removing the inner casing once slotted in. To address this, the second prototype incorporated handles to enable easier separation of the casing and hub, thus improving ease of use. However, the handle placement in the second prototype posed challenges, so a third prototype was developed. This final iteration brought the handles closer together for easier gripping and positioned them to eliminate the need for the user to reach across the central area of the pot.

4.2.2 Pot Attachment

Various methods to attach our product to pots were explored, but the most promising approach was inspired by lid sitters [1]. Initial prototypes were created through 3D printing, but these faced stability issues, with the attachment frequently falling to one side. We then redesigned the attachment with three contact points and planned to use a high-friction material, such as silicone, for better stability. To achieve this effect, we used TPU filament. However, using TPU for the entire attachment resulted in excessive flexion. Consequently, we revised the design to incorporate TPU only at the contact points, ensuring a stronger hold. The contact points were also modified to function like a clothes peg mechanism, providing a more secure grip. Finally, a simple detent mechanism was added to the joint allowing it to snap in place for easier use as they can set the angle of the arm and thus the radius of it before placing the device on the desired pot.

4.2.3 Stirring Attachment

As stated earlier in the concept development approach, we aimed for the stirring attachment to be a one-size-fits-all solution. This led to the initial ideation of a linkage system for the stirring attachment. The premise was that users would be able to adjust the shape to fit their pots and achieve different stirring mechanics. While this design maintained its form due to the linkage's stiffness (as shown in Figure 5), concerns arose regarding dirt accumulation in the small separations and the necessary size of these separations to ensure the linkage's stiffness at scale. Inspired by the concept of a pipe cleaner, we considered using a similar design, as pipe cleaners are easily mouldable. This led to the creation of a prototype using copper wire and plastic tubing, which confirmed the feasibility of a mouldable and cleanable design. This approach provided flexibility in adapting to various pot sizes and food consistencies, enhancing user convenience.

4.2.4 Central Rod Attachment

A basic electronic circuit was constructed using an Arduino Uno micro controller, a H-bridge, and a simple DC motor. Two different rod shapes were designed: one was purely cylindrical, while the other featured a spiral helical design with a pointed, rounded tip. We conducted tests for both rod attachments with the aforementioned electronic circuit to determine the optimal shape for the central rod connecting to the motor. The tests revealed that the central rod with the spiral helical design distributed spices more effectively. This design addresses a common issue with existing automatic stirrers where spices are not well-distributed, this will be crucial for enhancing the cooking experience for our elderly users.

With the understanding that the helical shape of the central rod better distributes spices, it was crucial to address other aspects such as the rod's fit into the hub and its rotation by the motor. To achieve this, a snap-fit design was developed, allowing the central

rod attachment shaft to slot securely into the hub. Additionally, to accommodate different pot depths, the central rod attachment shaft was designed to extend in length, akin to how modern umbrellas extend their shafts when the button is pressed. However, in this case, gravity facilitates the extension instead of a button. The bottom of the central rod attachment will be covered in silicone, ensuring it is food-safe and won't scrape the pot, making it suitable for consistent contact with the bottom of various pots.

4.2.5 Planetary Gears

For our stirrer to function efficiently, a relatively high torque was necessary. We decided to use a planetary gear system [2] and printed and tested these gears with our motor to ensure they produced enough friction to reduce speed and increase torque. The sun gear had to be printed multiple times due to the bore fitting needing to be perfect in order to allow enough friction to spin the rest of the gears within the planetary gear system.

4.3 Material selection

The main thing we have to worry about with our product's material selection is that it needs to be food and temperature safe. Because of this, the following materials have been chosen for the following parts:

- **ABS:** This is a versatile material which is both temperature resistant and, more importantly, food safe. This is why we would use it for the majority of the components ranging from the outside shells, to the inner base, the attachment's rod, the end tube covers, and the main bit of the arms. Its durable nature and ability to be injection moulded further adds to our desire to use it.
- **Silicone:** Silicone has been used for 4 components: To cover the stirrer as it's mouldable and food safe, to cover the support arm's joint so food and dirt doesn't get in them, for the grip on the support arms so they stick to the pot, and on the top to seal and cover the electronics while letting light through and show the button information.
- **Nylon:** The gears were planned to be made out of nylon due to it's self lubricating nature allowing us to avoid the need for grease and it's durable nature resulting in them being long lasting.
- **Copper:** Copper was used in the inside of the mouldable stirrer as it is what allows it to be bent but still keep its shape while cooking.

5 Electronic Component Selection

This section was especially important as making sure that the right specifications electronics were bought was essential. To find the different electronic components such as the motor specification and, the battery needed calculations were carried out in order find this.

5.1 Outputs

5.1.1 Motor Requirements

Based on our research, we determined that our device should be capable of at least stirring stews, with a maximum viscosity of approximately 2500 mPa·s [3]. The torque T required for a cylindrical rod rotating in a viscous fluid was approximated using the formula [4]:

$$T = 2\pi\eta L r^3 \omega \tag{1}$$

where:

- η is the dynamic viscosity (Pa·s or mPa·s/1000)
- L is the length of the stirring rod (m)
- r is the radius of the stirring rod (m)
- ω is the angular velocity (rad/s)

Given that the maximum viscosity stirring attachment had the following dimensions:

- $\eta = 2500\,\text{mPa} \cdot \text{s} = 2.5\,\text{Pa} \cdot \text{s}$
- $L = 0.8\,\text{m}$
- $r = 0.009\,\text{m}$
- $\omega = 10\,\text{rad/s}$

The torque required was calculated as follows and the value was found to be aproximately 1.83×10^{-4}, Nm To account for the gear system, we used the gear ratio R. The gear ratio for a planetary gear system is given by:

$$R = 1 + \frac{N_{\text{ring}}}{N_{\text{sun}}} \tag{2}$$

Given that our gear system had 15 teeth on the sun gear and 70 teeth on the ring gear, this resulted in

a gear ratio of approximately 5.67. To determine the torque required from the motor before gear reduction, we used the previously calculated torque needed to rotate the stew and divided it by the gear ratio.

$$\begin{aligned} T_{\text{motor}} &= \frac{T}{R} \\ &= \frac{1.83 \times 10^{-4}}{5.67} \\ &\approx 3.23 \times 10^{-5}\,\text{Nm} \end{aligned}$$

To ensure that the Cuttlesw!sh could operate efficiently, we needed a motor with a torque greater than 3.23×10^{-5} Nm. After researching various options, we selected a motor with a torque of 0.01052 Nm. This value significantly exceeded our minimum requirement, ensuring that the motor would meet the necessary performance criteria for our device.

5.1.2 Neopixel Ring

In the prototype phase, we opted against using a screen. Instead, we selected a 24-neopixel Adafruit LED ring as the primary user interface. This ring will display three modes, each represented by different colors, providing a clear and intuitive user experience.

5.1.3 Speaker

A speaker is included to alert users when the stirring process is complete. It will also play startup and shutdown sounds, indicating the device's boot-up and shut-down processes. Also to help users whilst switching to different modes a beep sound will be induced to signify moving to a new mode.

5.2 Inputs

5.2.1 Rotatory Encoder

A rotary encoder was chosen to allow the user to easily adjust the dial, thereby changing the Neopixel ring display and varying the motor speed. Unlike a potentiometer, a rotary encoder does not need to maintain a specific position, making it more suitable for our application as the rotatory does not have a dedicated start/stop.

5.2.2 Buttons

Two buttons will be implemented: one to power the system on and off, and another to switch between modes. The modes include temperature sensing, motor speed adjustment, and a timer. In order to start up the stirrer once a speed and the timer have been set users have to hold down the mode button for two seconds to start the stirrer. Capacitive touch was considered however was omitted due to the fear that tremors could trigger the sensor when not wanted.

5.2.3 Infrared Sensor

An infrared temperature sensor was selected to measure and display the temperature of the food being cooked via the LED ring. It was important to specifically acquire an infrared temperature sensor, because the temperature module would enable accurate temperature readings, ensuring optimal cooking results. It was also important that the infrared sensor selected could withstand high temperatures, due to the nature of the design it will be under hot conditions frequently.

5.3 Other Components

- **H-bridge:** A H-bridge is required to control the motor power
- **Charging Module:** This module is used to charge the device's battery.
- **Arduino Nano:** The micro-controller manages all components, ensuring seamless operation and integration.

5.4 Battery Calculations

To calculate the battery requirements, the current draw for each component was determined. This assumes a worst-case scenario where all components are simultaneously being used at their maximum rating. Although this is unlikely, it is necessary to ensure that the selected battery can meet the maximum required current draw. In order to do this specifications of the components were researched, and a power supply was utilised to check the current draw of each component at 5V.

To ensure that the Cuttlesw!sh can operate for 4

Table 1
Current draw of various components

Component	Current Draw (mA)
Arduino Nano	200
Infrared Sensor	15
NeoPixel Ring	432
Speaker	200
DC Motor	700
H-Bridge	100
Buttons	10
Encoder	50
Total	**1707**

hours continuously, the total current draw was multiplied by 4 to determine the required battery capacity. Finding a single battery that is both compact and fits the required specifications proved challenging. Therefore, we opted to use two 3.7V lithium batteries in conjunction with a step-up converter to increase the voltage to 5V. This setup meets the device's power needs while maintaining a compact form factor.

$$Battery, Life, (Hrs) = \frac{Battery, Capacity, (mAh)}{Load, Current, (mA)} \quad (3)$$

Given the total current draw of 1707 mA, for 4 hours of operation:

$$Battery, Capacity = 1.71, A \times 4, hours \quad = 6.8, Ah$$

Two 3.7V, 3400mAh lithium batteries connected in parallel would provide a total capacity of 6800mAh, meeting the requirement.

5.5 Electronic Circuit Design

The Figure below shows the final circuit and how each components were connected to achieve full functionality.

6 Design For Assembly

Assembly was at the forefront when designing the Cuttleswish. To begin, the design was centred around it, using snap fittings, with self-locating features allowing for quick and intuitive assembly.

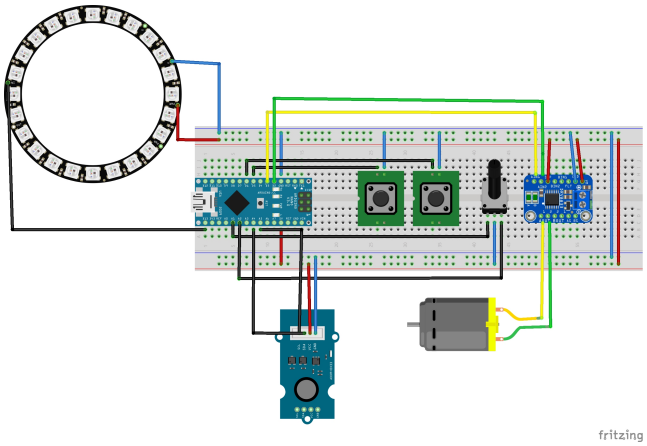


Figure 7. Finalised Electronic Circuit

6.1 Snap fittings

Multiple components employed snap fittings for fastening parts together, thereby allowing us to avoid adhesives and other permanent fastening methods. The first components to which snap fits were required to be included was the Attachment Rod. This was needed as the inner rod needed to facilitate linear motion to adjust the height, but still have the max height be limited so it wouldn't fall out. While the original design had the inner rod inserted in the top and would just have studs sticking out limiting the range of motion, the use of snap fits allowed for a much cleaner finished product, whose mechanism can be seen below:

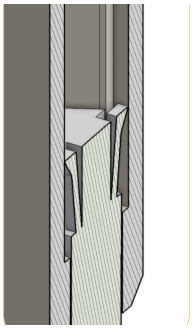


Figure 8. Snap fit on the attachment rod

Returning to the original design, it was determined that by making the top knob slot into place as well, we could use clamping to hold the entire design together and eliminate the need for more permanent fixing techniques. To do this, the Knob Extension had a grove cut into the middle which would slot into the spurs which were attached to the main electronics base, and the rotary encoder wheel:

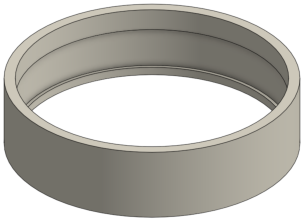


Figure 9. Groove in the knob extension

The last major components to use snap fits were the support arms, whose joints were thinned out over various iterations to allow for them to bend slightly, thus snapping into the detent mechanism without a need of a rod or multiple pieces.

6.2 Screws

The majority of the electronics which were not soldered would have to be fixed via screws; these being the Arduino Nano, the Motor, and the IR Sensor. However, some issues arose regarding space as the design focused on being compact, something that was seen in the version development resulting in the designs shrinking throughout the versions. The main issue was with the motor not having space for screw heads behind it but a quick solution for this was found through the use of socket set grub screws which are headless and therefore avoid wasting precious space.

The other main components requiring screws were the start of the support arms which were attached to the main shell. These had 2 M2 screws per component fixing them to the shell via threading allowing for simple replacement of the components if needed as the detent mechanism is likely to wear out quickly.

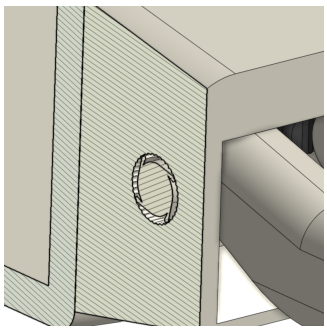


Figure 10. Detent Mechanism in the Arm Joint

6.3 Provisioning For Assembly

Three key ideas were kept in mind throughout the designing and prototyping to allow the different components to fit together:

- **Clearance Fits:** These are spaces between mating parts which needed some free movement to reduce friction, such as in the case of the planet gears, or to allow smoother movement and user interaction, such as in the case of the tolerance within the bayonet mechanism on the top attachment which allowed for smoother assembly , or in the case of the space between the outside shell and the inner casing, to allow for easier extraction for the user.
- **Interference Fits:** These were crucial to the design to ensure that the electronic components would be sealed from the potential heat and steam that would arise from the food. For it, we used a bearing to seal the hub's casing to the ring gear's bayonet mechanism, whilst the infrared plugged up the other lower hole. Then, looking at the opposite end of the Cuttleswish, we sealed the top using a thin silicone sheet which ensure complete protection to the electronics within.
- **Allowance & Enveloping:** Some components, especially around moving parts, had to have additional space allocated to them to ensure that no damage was done to both them and the moving part. The most glaring example, that we found while prototyping, was the IR Sensor whose pins were scraping against the bottom of the ring gear, thereby resulting in us decreasing the thickness of the gear.

7 Design For Manufacturing

In order to both ensure that our product could fill the brief and thus be able to be manufactured effectively and with industrial methods. This was ensured by creating parts that could be injection moulded as much as possible, or could also be stamped for certain components for quicker manufacture.

7.1 Injection Moulding

Components like the casings and the arms have been designed to be injection moulded to ensure lower man-

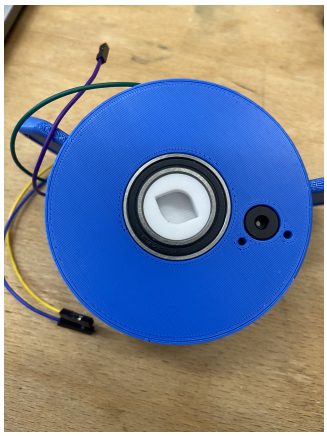


Figure 11. Seal on Bearing and IR Sensor

ufacture costs and high production rates. Although for our models they were 3D printed, the final product was still designed with manufacture in mind resulting in simpler shapes where possible, with snap fittings for assembly.

7.2 Stamping

Components like the silicone cover had to be stamped out of silicone sheets to ensure that they would both be cut to the correct shape through the use of a circular die and high pressure, meanwhile, the numbers and icons could be included in the die to allow them to embossed into the material for the user to then see.

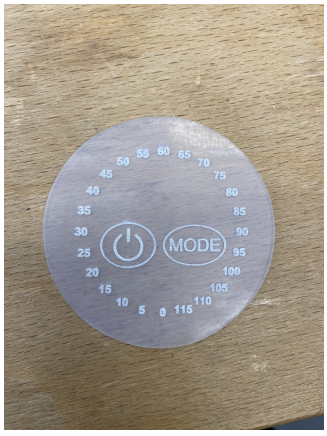


Figure 12. Silicone cover

7.3 Extrusion

Two pieces would have to be extruded within our design. One is the copper rod that is used to control the shape of the mixing tentacles, and the other is the silicone cover for it. Both would use dies and extrusion for production, however, these are the type of components that are easier to outsource production for to specialised companies, just as we have done.

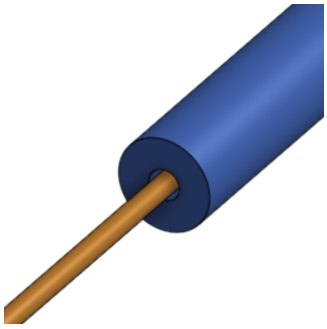


Figure 13. Copper rod within silicone cover

8 Branding Consideration

In order to select the brand that best aligns with our product motivation, helping elderly cook, and aesthetics, a series of different brands were considered.

8.1 Benchmarking Analysis: Product

8.1.1 Uutensil Stirr Automatic Pan Stirrer

The Uutensil Stirr is a cheap, simple setup, automatic stirrer, featuring detachable, dishwasher-safe legs made of food-grade plastic. It offers three speed settings, but the battery life is limited to about four hours.



Figure 14. Uutensil Stirr Automatic Pan Stirrer

8.1.2 Gourmia Stirrific



Figure 15. Gourmia Stirrific

The Gourmia Stiriffic is a versatile automatic stirrer with multiple speed options, allowing it to stir a wide variety of dishes quietly. Its adjustable clip design fits almost all pots and pans.

8.1.3 Ardente Gourmet Stirrer

The Ardente Gourmet Stirrer features adjustable handles that attach to the pot's sides, providing stability. It has a single-speed motor, ideal for slow stirring of soups and stews. However, the lack of multiple speed settings might limit its versatility for different types of cooking.



Figure 16. Ardente Gourmet Stirrer

8.1.4 StirMate Smart Pot Stirrer



Figure 17. StirMate Smart Pot Stirrer

The StirMate Smart Pot Stirrer is an automatic stirrer where inspiration was drawn for the original design, with adjustable arms to fit various pot sizes. It features a powerful motor and rechargeable battery, making it suitable for continuous use.

8.2 Benchmarking Analysis: Brand

8.2.1 Kitchenaid

Kitchenaid is known for its stylish and sleek finish, often associated with high-quality kitchen appliances. Their product line includes kitchen mixers that are both functional and visually appealing, making them a strong contender for our design inspiration.

8.2.2 Joseph Joseph

Joseph Joseph specialises in products that prioritise ease of use and simplicity. Their minimalist aesthetic, combined with a splash of colour, caters to a wide range of users, particularly those seeking practical yet visually engaging kitchen tools.

8.2.3 Ototo

Ototo is renowned for its creative and quirky kitchen gadgets. Their playful designs add a touch of fun to everyday cooking tasks, making kitchen activities more enjoyable. This approach to blending functionality with a lighthearted aesthetic makes Ototo an appealing reference for our project.

8.2.4 Peleg Design

Peleg Design stands out for its playful element and practical kitchen tools. Known for their whimsical designs and pun-based names, Peleg Design offers a unique aesthetic that combines functionality with a sense of fun, making it an ideal choice for our product.

8.3 Brand Selection

We selected Peleg Design as our source of inspiration for the aforementioned brands. Their emphasis on playful and practical kitchen tools aligns perfectly with our goal of creating a user-friendly and engaging product. Peleg Design is the ideal match for our Cuttleswish™ because of its unique blend of zoological elements, functional practicality, and whimsical design elements.

9 Brand Implementations

9.1 Brand Implementations on the Physical Product

The following modifications were made to the physical features of the product inspired by Peleg Design:

9.1.1 Functional Aesthetics

Peleg Design products are known for combining functionality with aesthetic appeal. For the Cuttleswish™, we aim to incorporate sleek, visually appealing designs that are also highly functional in a kitchen environment. This includes smooth finishes, ergonomic handles, and a modern, minimalist look that fits well with contemporary kitchen decor.

9.1.2 Humour and Whimsy

A key aspect of Peleg Design's products is the element of humor and whimsy. We plan to integrate playful elements into the Cuttleswish™ by using bright colors and fun shapes inspired by marine life. This not only makes the product more enjoyable to use but also

adds a unique touch that differentiates it from standard kitchen tools.

9.2 Brand Implementations on the Product's Values

The values of Peleg Design influence how we market and develop the Cuttleswish™:

9.2.1 Functionality

Peleg Design emphasises the importance of functionality in their products. For Cuttleswish™, this translates to ensuring that the product is easy to use, efficient, and meets the needs of our target market. Features such as adjustable stirring speeds and detachable, dishwasher-safe components are essential to maintaining high functionality.

9.2.2 Aesthetics

The aesthetic appeal of a product is crucial to Peleg Design. We aim to make Cuttleswish™ not only a tool but also a decorative item that users are proud to display in their kitchens. This includes using high-quality materials and a design that complements modern kitchen interiors.

9.2.3 Humour

Incorporating humour into the design helps to create a more engaging user experience. By adding whimsical elements inspired by Peleg Design, such as fun shapes and bright colours, we aim to make the Cuttleswish™ a delightful addition to any kitchen, encouraging more frequent and enjoyable use.

10 Product Compliance: Research

Peleg Design take pride in their products, which are now shipping worldwide. The Cuttleswish™ being distributed in the UK specifically, it is essential that the product adheres to the appropriate UK and EU-specific standards and legislation for both legality and customer safety.

10.1 UKCA

The UKCA (UK Conformity Assessed) marking is essential for products sold in Great Britain to ensure they meet UK regulations and to replace the CE mark following Brexit. Until the end of 2027, the UKCA mark can be placed on a label or accompanying document. After this period, the marking must be directly affixed to the product itself. [5]



Figure 18. UKCA Marking

10.2 CE

The CE marking certifies that a product complies with EU safety, health, and environmental standards, allowing free movement within the European Economic Area. It must be visibly and legibly affixed to the product, its packaging, or accompanying documents.



Figure 19. CE Marking

10.3 Marking Requirements

In order to obtain these markings for the product, the following steps must be taken:

- Conduct a **Conformity Assessment**.
- Maintain a **Technical File** containing: a product description and specifications, design and manufacturing drawings, risk assessments, test reports, and a user manual with safety information. This is to be kept 10 years after the product sees the market.
- Draft a **Declaration of Conformity** containing: manufacturers name and address, product identification, list of relevant EU directives and standards and declaration of compliance.

10.4 General Legislation

The current general EU legislation were studied and we selected those which applied to our product, detailed below.

RoHS (Restriction of Hazardous Substances in Electrical and Electronic Equipment)

This is the directive on the restriction of the use of specific hazardous substances in electrical and electronic equipment. This ensures the Cuttleswish™ is safe for users and environmentally friendly.

EMC/EMF (Electromagnetic Compatibility and Electromagnetic Fields)

Refers to the capacity of electrical equipment and systems to operate effectively within their electromagnetic surroundings. This involves managing any subliminal electromagnetic interference (EMI) between the Cuttleswish™ and other kitchen appliances.

WEEE (Waste from Electrical and Electronic Equipment)

This details regulations and guidelines to ensure the safe disposal and recycling of the waste electronics of the CuttleswishTM at the end of life, minimising environmental impact.

EcoDesign Directive

This establishes strict ecological standards for energy-consuming items, guaranteeing the energy efficiency and low environmental impact of the Cuttleswish™ for the course of its lifetime.

Battery Directive

This aims to minimise the negative impact of batteries on the environment. Overall, ensuring that the rechargeable batteries used in the Cuttleswish™ are safe, environmentally friendly, and correctly labelled for recycling and disposal.

Food Contact Materials Regulation

This regulation ensures that materials intended to come into contact with food such as the Cuttleswish™ stirring attachment and shaft are safe and do not release harmful substances, protecting consumer health.

11 Product Compliance: Conformity

In order to ensure the highest level of quality and safety for the Cuttleswish™, we strictly adhere to all relevant industry standards. Complete legal compliance and improved product performance are guaranteed by these standards, which provide comprehensive technical guidelines that satisfy both consumer expectations and regulatory requirements.

11.1 RoHS (Restriction of Hazardous Substances in Electrical and Electronic Equipment)

UKCA Legislation: The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (SI 2012/3032)

CE Legislation: RoHS Directive 2011/65/EU

- Standard: EN 50581 (Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances)

11.2 EMC/EMF (Electromagnetic Compatibility and Electromagnetic Fields)

UKCA Legislation: Electromagnetic Compatibility Regulations 2016 (SI 2016/1091)

CE Legislation: Electromagnetic Compatibility Directive 2014/30/EU (EMCD 2014/30/EU)

- Standard: EN 55032 (Electromagnetic compatibility of multimedia equipment - Emission Requirements)
- Standard: EN 61000-3-2 (Electromagnetic compatibility (EMC) - Limits for harmonic current emissions)
- Standard: EN 61000-3-3 (Electromagnetic compatibility (EMC) - Limitation of voltage changes, voltage fluctuations and flicker)

11.3 WEEE (Waste from Electrical and Electronic Equipment)

UKCA Legislation: The Waste Electrical and Electronic Equipment Regulations 2013 (SI 2013/3113)

CE Legislation: Waste Electrical and Electronic Equipment Directive 2012/19/EU (WEEE 2012/19/EU)

- Standard: EN 50625 (Collection, logistics & treatment requirements for WEEE)

11.4 EcoDesign Directive

UKCA Legislation: The Ecodesign for Energy-Related Products Regulations 2010 (SI 2010/2617)

CE Legislation: Eco-design Directive 2009/125/EC

- Standard: EN 50564 (Electrical and electronic household and office equipment - Measurement of low power consumption)
- Standard: EN 60960-1 (Requirements for the reduction of hazardous substances in multimedia equipment)

11.5 Battery Directive

UKCA Legislation: The Batteries and Accumulators (Placing on the Market) Regulations 2008 (SI 2008/2164)

CE Legislation: Battery Directive 2006/66/EC

- Standard: EN 62133 (Secondary cells and batteries containing alkaline or other non-acid electrolytes - Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications)
- Standard: IEC 60086-4 (Primary batteries - Safety of lithium batteries)

11.6 Food Contact Materials Regulation

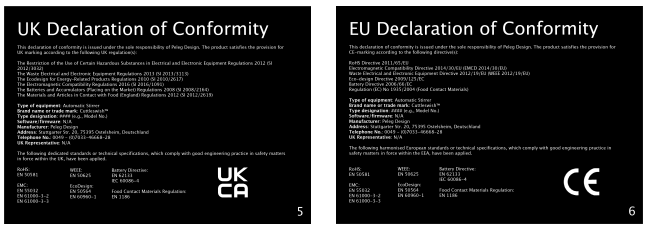
UKCA Legislation: The Materials and Articles in Contact with Food (England) Regulations 2012 (SI 2012/2619)

CE Legislation: Regulation (EC) No 1935/2004

- Standard: EN 1186 (Materials and articles in contact with foodstuffs - Plastics)

11.7 Declaration

A declaration of conformity must be sent with the product in order to obtain and display the UKCA and CE marks, and the marks must be clearly attached to the product.



(a) UK Declaration of Conformity

(b) EU Declaration of Conformity

Figure 20. Declarations of Conformity

12 Labelling and User Guide

12.1 Product Labels

12.1.1 WEEE Label

The Waste Electrical and Electronic Equipment (WEEE) directive aims to minimise the environmental impact of electronic waste. Users are advised to properly recycle the product instead of disposing of it in general waste.

12.1.2 UKCA and CE Labels

As stated previously, the UKCA (UK Conformity Assessed) and CE marks must be affixed to the product in a visible and legible manner at least 5mm in height.

12.1.3 Food Contact Material Label

The food-safe symbol indicates that the materials used in the product are safe around food and comply with harmful substance regulations.



(a) WEEE Marking

(b) UKCA and CE Marking

(c) Food Safe Marking

Figure 21. Product Label Markings

12.2 User Guide

Given our target market, it is imperative we ensure that the pamphlet contains clear instructions such that the user has full confidence when using their Cuttleswish™. The pamphlet is comprised of the following pages.

12.2.1 Front and Back Cover

The covers of the pamphlet displays the product name on the front and to corresponding company on the back. This is complete with a background of the Cuttleswish™ motif tessellation.

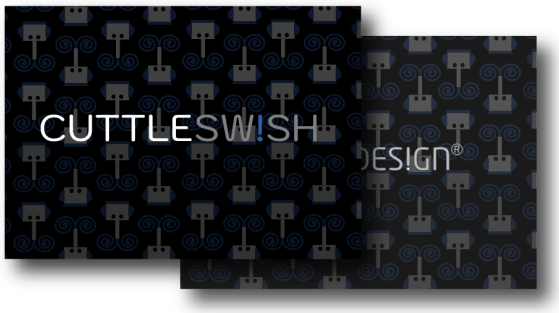


Figure 22. Pamphlet - Front and Back Cover

12.2.2 Instructions

This section details the step-by-step instructions for set up and usage of the Cuttleswish™. It guarantees that users can learn how to use the device safely and effectively in no time.

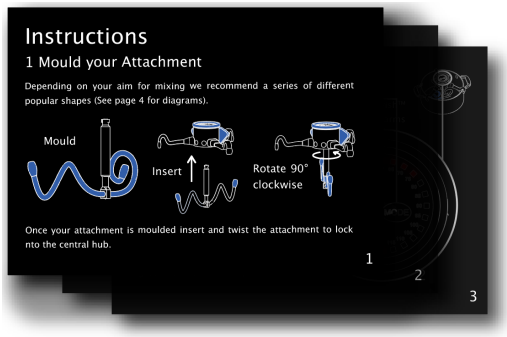


Figure 23. Pamphlet - Instructions

12.2.3 Attachment Shapes

This page showcases various attachments of the Cuttleswish™, each with a name and description of its stirring motion. This helps users select the appropriate attachment for their cooking needs. Elements from multiple attachment shapes can be incorporated into one to combine different attributes.

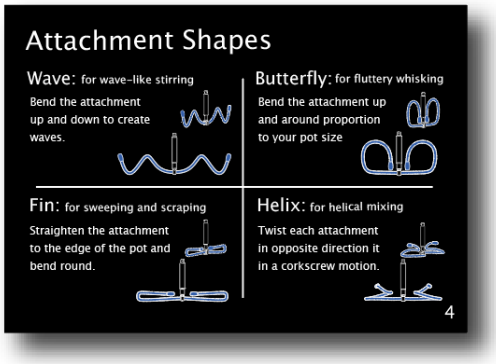


Figure 24. Pamphlet - Attachment Shapes

12.2.4 Safety

This section outlines the safety instructions and warnings necessary for the safe operation of the Cuttleswish™, in order to prevent accidents and guarantee user safety.



Figure 25. Pamphlet - Safety

12.2.5 Maintenance and Support

Maintenance tips and support information are provided here. Users can learn how to maintain their Cuttleswish™ and find contact information for customer support, in addition to the digital version of the manual to display on a larger screen to accommodate those with visual impairments.



Figure 26. Pamphlet - Maintenance and Support

12.2.6 Declaration of Conformity

The final pages include the Declaration of Conformity: the legal documentation confirming that the Cuttleswish™ complies with the relevant UKCA and CE regulations.



Figure 27. Pamphlet - Declaration of Conformity

13 Product Packaging Design

13.1 Branding and Inspiration

Analysing Peleg Design's pre-existing packaging and their visual identity, we aimed to take inspiration from this and implement the core features into our design. These mainly include the front hero image, labelled with the product name, description and company logo, in addition to the sleek black aesthetic.

13.2 Target Market

Peleg Design from the style of their products currently pitch them to the 20 to 50 demographic and this is communicated in their packaging. With our target market being elderly users we have taken priority in making the packaging and product easily accessible. This firstly means making all font size a minimum of 12pt as deteriorating eyesight in our demographic is common.
Additionally the use of info-graphics in combination with simple phrasing and explanations, allows users to easily comprehend and familiarise themselves with operating the product. This can be highlighted on the condensed manual instructions found on the side of the packaging, in addition to QR code for an enlarge more thorough set of instructions.

13.3 Functionality and Design

The key aims of the product packaging are first to entice consumers, provide quick product identification and help them realise their need for the product, and finally, provide suitable protection and compact functionality.

Triangular Box

To entice the user in combination we chose a triangular box, this silhouette noticeably distinguishes the Cuttleswish™ from other products in store and catches the attention of consumers. This shape also reduces packaging protection by 50% which would otherwise be wasted material.

Brief Instruction

Failing that the user reads the pamphlet or misplaces it a condensed set of instructions complete with infographics can be found on the side of the packaging. These three steps give a short guide to highlight the accessibility and ease of use of the Cuttleswish™.

Support

Turning the packaging you are met with a more detailed description of the product, in addition to customer support information and a QR for an enlarged version of the pamphlet and user guide.

13.4 Internal Design

The packaging is comprised of two internal layers. Once opened the first is revealed housing the Cut-

tleswish™ fully assembled, bar the attachment creating a "plug 'n' play" experience for the user with minimal setup needed.
Once the Cuttle swish and upper layer are removed, the second layer is revealed, housing the charging components, stirring attachment and pamphlet.

13.5 Prototyping

The packaging net was designed in Adobe Illustrator allowing precise text and image refinement. This net was then converted to a DXF file with the upper triangle fulfilling a diameter and of 390mm and a height of 135mm complete with 10mm tabs to construct the net to a shell. This design was then projected on to our model to view the packaging in situ.

13.6 Labels

The following labels have be see on the bottom shell of the packaging to show that the product to indicate material properties and dictate disposal guidelines for the packaging.

The Widely Recycled Label

This label dictates that the packaging can be collected and recycled by the majority of local authorities in the UK. It ensures that the materials used for Cuttleswish™ packaging are widely accepted for recycling.

The Forest Stewardship Council (FSC) Label

This symbol shows that the cardboard or paper packaging comes from forests that are managed responsibly and sustainably. It certifies that the packaging material for Cuttleswish™ is environmentally friendly and sourced from well-managed forests.

The Green Dot

This label is used as an EU to signify that the manufacturer has financially contributed to the packaging material recovery and recycling. It confirms that Cuttleswish™ packaging complies with EU recycling regulations and supports environmental sustainability.

13.6.1 The Mobius Loop (20 PAP)

The Mobius Loop symbol indicates that the packaging is recyclable. The number 20 and the letters "PAP" specifies the material, being corrugated cardboard. This similarly to the green dot ensures the Cuttleswish™ packaging is disposed responsibly.



(a) Widely Recycled Marking

(b) FSC Marking

(c) Green Dot Marking

(d) Mobius Loop Marking

Figure 28. Packaging Label Markings

13.7 Material Choice

The material of choice for the packaging is corrugated cardboard. This can be sustainably sourced at low cost, being gloss laminated to provide a sleek and elegant finish. Two layers of packaging foam with female cutouts will then be used to house the cable, plug, pamphlet and Cuttleswish™ hub and shell to provide suitable protection in transit between the manufacturer and consumer.

14 Final Design - Features



Figure 29. Render of finished product

14.1 Tactile Rotary Control

The key to any good design is the ease of use, especially when working with the elderly. Because of this,

we focused in initial development our desired inputs and outputs. A design choice we selected as an input was the large rotary wheel which controls both the motor speed and timer function. To change mode the user would simply need to press a button, and then hold it down for 2 seconds to begin swishing.

14.2 Visual and Auditory Feedback

The simple nature of the product was further developed through the use of only two feedback outputs, which worked in conjunction to give the user all the information they required. Starting with the visual component, we use a Neopixel LED Ring to show different information depending on the mode: When setting the timer, the ring colours green and has a scale of 5 minutes for each LED, with each minute in between causing the final LED to decrease in brightness. Secondly, we can control the motor speed, a blue light array, with final mode, a red light array, displaying temperature also on the same 5°scale as the timer, allowing us to have a single text scale dictate them both. A speaker is also featured for key auditory feedback. This being boot-up and boot-down sounds, alerts for a changed mode, and a final alarm when the timer is finished.

14.3 Adjustable Sizes

To make our device able to fit a range of pot diameters, we need it to be adjustable. The use of folding support arms with fixable TPU ends allowed us to adjust to multiple sized pots by changing the radius. At the same time, the height of the attachment adjusts automatically to the depth of the pot through the use of the weight of the attachment and gravity, avoiding the use of manual intervention when ensuring proper fit.

14.4 Mouldable Stirrer

The final challenge we tackled was the ability for the user to adjust to different pot shapes, and the needs of different mixing methods, which can range from having large solids needing to be moved around, to the bottom of the pan needing to be scraped. Having the tentacle mixer mouldable allows the user to adjust to whatever shape of pot and pan, whether it be a deep dish, a stock pot, or even a wok.



Figure 30. Finished product

15 Business Case

The Cuttleswish is a product designed to bring joy, health, and social interaction to cooking. However, this is something that the elderly often struggle with through both physical and cognitive challenges which remove their ability to enjoy cooking. Our product offers users an adaptable and easy to use device that will bring back the enjoyment they had with cooking and remove many of the struggles and challenges.

15.1 The Market

The market within the UK is constantly growing as life expectancy increases and the population ages more and more, with a 2021 Census stating that there were over 15.5 million people in the UK aged over 60 [6]. This directly results in a growing need for devices which aid daily activities for the elderly, especially key ones like cooking, allowing them to live an independent lifestyle. Furthermore, the more approachable and friendly look of the design brings it that one step above its competitors.

15.2 Product Feasibility, Desirability And Viability

- **Feasibility**: Given the simple functionality and simple material choice, the project is completely doable to be manufactured and sold by PELEGDESIGN.

- **Desirability**: Looking at the growing elderly population, and the high demand for devices that aid them in remaining more independent, this product has a solid market base.
- **Viability**: The lack of expensive components and materials, together with the desirability of the product, makes this a compelling business opportunity that is sure to lead to revenue.

15.3 Revenue Model

The primary revenue for the Cuttleswish will be generated through product sales, especially focused on Amazon as a supplier which would allow us to reach a larger market with focus on elderly and carers for the elderly. We could also expand and create partnerships with senior living facilities and healthcare providers. Furthermore, we can even market this to people with disabilities and injuries which might struggle with cooking as there is nothing stopping us from expanding our aims.

16 Object Validation

To ensure that our project would meet the objectives that were put forward at the start through our brief, testing of the final product was conducted both individually to test the mechanical and electronic functions, and with elderly users to gain feedback.

16.1 Mechanical & Electronic Functions

16.1.1 Controlling the device

Running the device allowed to test user functions ensured that it worked properly and met our desired specifications. The electronics were integrated properly ensuring that the desired inputs led to the desired outputs being displayed, with the only small issue arising being the noise from the gears. However, that could be fixed by both greasing the gears, and having tighter tolerances for them in the final design.

16.1.2 Adjustable Dimensions

As our device needed to be adjustable to many different pots, we had the outside arms bend to be able to change the radius, however, while this worked well for very large pots, as the size decreased, the system struggled to fit on the smaller range of pots below 22cm. However, the height adjusting system worked very well for our needs.

16.1.3 Gripping the pot

Something that we had envisioned early on but weren't able to test until almost the final prototypes was the method to grip onto the pots. Many were skeptical on its ability to grip, however, the final design proved that it would work extremely well.

16.2 Feedback

16.3 Elena Sachetti

With an overall positive review of the product, Elena really liked the look of the product, even if she described it at one point as looking like "a man peeking out of a dumpster". The thing that struck her the most though was the customizability of the design, allowing you to adapt to differently sized pots, and even more, the ability to change the shape of the stirrer.

16.4 Anthony Walsh

Anthony was really excited about seeing this product come to fruition, having been part of this from the start, and having been constantly updated on the development. He really liked it's appearance, describing it as a "friendly critter", and was even more amazed with the ease of use.

17 Project Plan Gantt Chart

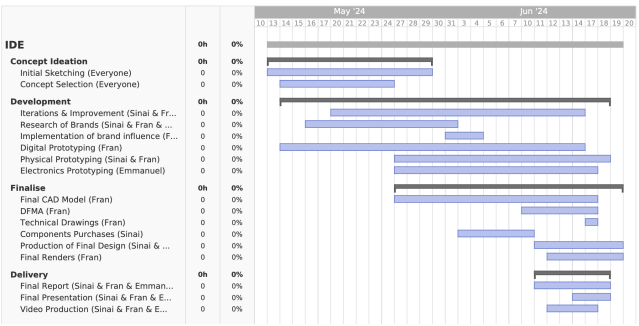


Figure 31. Gantt Chart showing the project plan

18 Team Roles & Contributions

18.1 Emmanuel - Chief Information Officer

Emmanuel focused on the electronics and writing:

- **Electronics**: When looking at the electronics, he did almost the whole programming for the group thereby ensuring that we got the project working as effectively as possible and to a high standard.
- **Calculations and Proof**: To ensure that the theory would match the practical aspects, Emmanuel spent a lot of time researching and calculating any proof that we'd need to ensure that our product would work.
- **Report & Portfolio**: Emmanuel led the group in the writing of the report structuring and formatting it.
- **Initial Prototyping**: He led forward the initial testing of ideas that we had to improve the mixing process.

18.2 Sinai- Chief Operations Officer

Sinai focused on user experience and the overall manufacture of the product:

- **Packaging**: Sinai was the key team member when designing and manufacturing the packaging for the product, running the whole process and ending up with an amazing finished product.
- **Label & Compliance**: He conducted the whole research around any safety and compliance that would be required for the product.
- **User Guide**: Sinai created a user guide for the product that would give users a whole range of instructions on use.
- **Soldering**: When it came to soldering the whole circuit together, he did an exceptional job ensuring the circuit worked.
- **Manufacture**: Through 3D printing, painting and assembly, Sinai ensured that the product was built to a high standard and with a quality finish.
- **Report & Portfolio**: Sinai wrote other large parts of the report, focusing mostly on product compliance and labels.

18.3 Francesco - Chief Technical Officer

Francesco was focused mostly on the digital development of the product:

- **Ideation & Iterations**: He created all 5 versions and had multiple innovative ideas on how to improve the design and new unusual methods to get experimental ideas to work.
- **CAD Development**: Francesco took the lead designing and ideating all 5 version's CAD models, including the final one, testing ideas and component design throughout.
- **Renders**: For this, Blender was use with virtual environments to give an optimal finish and allow for multiple views and locations for the product.
- **Video Production**: The majority of the video was also created on Blender with the aim of having an introduction to the video and exploded views. The rest of the video was filmed when the product was finally finished.
- **Report & Portfolio**: In the report, Francesco was mainly focused around the DFMA, the final design and a few other pages.

18.4 Alex - Chief Creative Officer

Alex only contributed the following things at the very beginning of the product:

- **Initial Ideation**: Alex contributed slightly in the original ideas for the design.
- **IR Sensor & Speaker**: Alex got the IR Sensor to almost work and created the basis of the code for the speaker.

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